INCH-POUND

MIL-A-81403(USAF)
AMENDMENT 3
17 JUNE 1991
SUPERSEDING
AMENDMENT 2
18 FEBRUARY 1986

MILITARY SPECIFICATION

ALTIMETER-ENCODER AAU-21/A

This amendment forms a part of MIL-A-81403(USAF), dated 15 June 1967, and is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1.1 Add title: Scope.

Add new paragraph:

"2.1 Government documents."

Delete existing paragraph number 2.1 and add: "2.1.1 Specifications, Standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed on the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

* Under SPECIFICATIONS Federal, add: "PPP-B-601 Boxes, Wood, Cleated Plywood", "PPP-B-636 Box, Shipping Fiberboard" and "PPP-B-1672 Boxes, Shipping, Reusable with Cushioning."

Under SPECIFICATION, Military, delete "MIL-E-5272", "MIL-D-1000/1" and "MIL-N-18307" entirely.

Add boxed paragraph to bottom center of page:

"Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/TIMLA, Tinker AFB OK 73145-5990 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter."

* Under STANDARDS, Military, delete "MIL-STD-143" and "MIL-STD-794" entirely. Add "MIL-STD-100, Engineering Drawing Practices"; "MIL-STD-129, Marking For Shipment And Storage"; "MIL-STD-130, Identification Marking Of U.S. Military Property"; "MIL-STD-810, Environmental Test Methods And Engineering Guidelines"; "MIL-STD-970, Specification And Standards, Order of Precedence For The Selection Of"; and "MIL-STD-2073/1A, DOD Materiel Procedures For Development And Application Of Packaging Requirements."

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* Add the following source statement:

"(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Ave, Building #4, Section D, Philadelphia, PA 19111-5094."

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* Add: "2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained."

3.1 Delete and add:

"3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.2.c) in accordance with 4.2."

* 3.2 Line 3, delete "MIL-STD-143" and add "MIL-STD-970."

* Add: "3.3.1.1 Reclamed materials. The use of reclaimed materials shall be encouraged to the maximum extent possible.

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3.11 In first line delete "MIL-N-18307" and add "MIL-STD-130."

3.14 In last line delete "MIL-D-1000/1" and add "MIL-STD-100."

* Delete "4.1" entirely and add:

"4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements."

* Add: "4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, hoever, this does not authorize submission of know defective material, either indicated or actual, nor does it commit the Government to accept defective material."

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4. Delete all references to "Qualification" and add "First article", except in 4.2.(b).

4.2.(b) Delete "Qualification conformance inspection" and add "Quality conformance inspection."

4.4.2 Delete and add:

"4.4.2 Sampling Plan B sample selection. Three samples shall be selected at random from the first 15 produced on contract."

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4.7.2 Line 3, delete "procedures of MIL-E-5272" and add "test method and procedure of MIL-STD-810."

4.7.2.2 In line two delete "Procedure II" and add "Method 502.1 Procedure I, except that in Step 2 the storage temperature shall be (-62°C) -30°F and the soak period shall be 72 hours. The operating temperature in Step 4 shall be (-54°C) -65°F and the stabilization period shall be 24 hours."

4.7.2.3 In line two delete "Procedure I" and add "Method 501.1 Procedure I, except delete Step 4."

4.7.2.7 Delete and add:

"4.7.2.7 Vibration failure. The Altimeter-Encoder shall be mounted on the vibration apparatus in its normal operation position. While being operated at one of the asterisked test points in Table II, shall be vibrated with circular motion of .018 to .020 inch diameter in a plane inclined 45 degrees to the horizontal plane and the frequency of vibration shall be varied uniformly from 5 to 50 Hertz and return once each hour for a three hour period."

4.7.2.8 In line two delete "Procedure I" and add "Method 513.2, Procedure II." To the end of the paragraph add: "During this test the Altimeter-Encoder shall be operated at on of the asterisked test points in Table II. There shall be no malfunction of the Altimeter-Encoder or power warning circuits during applied acceleration."

4.7.2.9 In line two delete "Procedure V, operational shock only" and add "Method 516.2-2. Amplitude of 15 g's peak acceleration, duration of 11±1 milli-seconds. During this test Altimeter-Encoder shall be installed in its normal operating position, with external lines installed, but not be operational."

4.7.2.13 In line two delete "Procedure I" and add "Method 507.1. Procedure I, except that the chamber temperature of Step 2 shall be (71°C) 160°F and the relative humidity of Step 4 shall be 95 percent."

4.7.2.14 In lines one and two delete "Procedure I" and add "Method 508.1, Procedure I."

4.7.2.15 In line two delete "Procedure I" and add "Method 509.1, Procedure 1."

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4.7.2.16 In line two delete "Procedure I" and add "Method 510.1, Procedure I, except that the air velocity in Step 1 shall be 300±200 feet per minute. The air temperature in Step 2 shall be (71°C) 160°F and the exposure period shall be six hours. Delete Step 3."

Add "4.9 Inspection and test. Tests of methods of preservation and packaging shall be accomplished in accordance with Section 4 of MIL-P-116 to ensure compliance with Section 5 of this specification."

* Delete section "5" entirely and add:

"5. PACKAGING

5.1 Preservation. Preservation shall be Level A or C as specified (see 6.2).

5.1.1 Preservation-Packing. Unless otherwise specified by the contracting activity, item shall be packaged in Quantity Unit Pack (QUP) of one each. Each item will be provided a preservation method 1A-8 IAW MIL-P-116.

5.1.2 Cleaning. Item shall be cleaned IAW MIL-P-116 1A-8 process.

5.1.3 Drying. Immediately after cleaning, the item shall be dried following any one or combination of the drying procedures listed in MIL-P-116. The drying procedures employed shall not be injurious to the item.

5.1.4 Level A. Item shall be preserved IAW MIL-P-116 and MIL-STD-2073/1A to provide a method 1A-8.

5.1.5 Level C. Item shall be individually preserved in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from the supply source to the first receiving activity as specified in MIL-P-116 and MIL-STD-2073/1A.

5.2 Packing. Packing shall be Level A, B, or C as specified (see 6.2).

5.2.1 Container-Cushioning. Intermediate container requirements shall conform to PPP-B-636 or PPP-B-1672, weather resistant, unless otherwise specified by the contracting activity. Intermediate containers shall be large enough to allow for application of sufficient cushioning material shall be of sufficient density and thickness to ensure shock transmission does not exceed peak values in G's established for the item.

5.2.2 Level A. Item will be preserved as specified in 5.1 and shall be packed in exterior containers conforming to PPP-B-636 or PPP-B-1672, weather resistant, unless otherwise specified by the contracting activity. Exterior container shall be uniform shape, size, and minimum tare and cube consistent with the protection required. Closure shall be in accordance with appropriate PPP-B-636 or PPP-B-1672 procedures as specified by contractor, Special Packaging Instructions (SPI), and contracting activity.

5.2.3 Level B. Same as Level "A" unless otherwise specified by Special Packaging Instructions (SPI) or the contracting activity.

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5.2.4 Level C. Item will be packed in containers in such a manner as to afford adequate protection against physical damage during direct domestic shipment from the supply source to the first receiving activity. These packs shall conform to MIL-STD-2073/1A.

5.3 Marking. Unit, intermediate, and exterior containers shall be marked IAW special markings required by the contractor, Special Packaging Instructions (SPI), and MIL-STD-129.

5.4 Inspection and Test. Test of methods of preservation shall be accomplished in accordance with Section 4 of MIL-P-116 to ensure compliance with Section 5 of this specification. Packaging tests shall be conducted IAW rough handling as specified in Federal Test Method Standard No. 101."

* Section 6 under NOTES add:

"(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)"

6.2(c) Delete "Reliability Qualification" and add "Reliability First Article."

* Add: "6.2(h) Issue of DODISS to be citedd in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1)."

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Delete "6.4" entirely and add new: "6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first article production items, a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.2."

Add: "6.6 Certain provision of this document are subject to international standardization agreement AIR-STD 10/60D. When amendment, revision, or cancellation of this document is proposed which will modify the international agreement concerned, the preparing activity takes appropriate action through internatinal standardization channels including the Departmental Standardization Offices (DepSo) to change the agreement or make other appropriate accomodations.

* Add: "6.7 The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content previous amendment."

Custodian:
AIR FORCE - 99

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